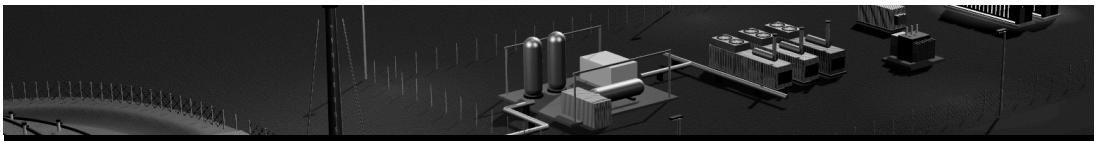
Albany (Rapp Road) Landfill, Albany, New York: site sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

2.1 MW	2	7 ac	188k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $563k	$7k to $15k	$7k to $578k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=758&CH4=48.8

THE MATH

Landfill gas flared beside the engine plant (2024)	758 scfm
Methane content (2024)	48.8%
Energy in the gas	22.6 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 2.1 MW
Compute (power less cooling and losses, /1.2)	about 1.7 MW
Compute modules (about 1 MW each)	2
Gas engines (running plus one spare)	3 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 7 acres
Gas bought per year (95% run time)	188k MMBtu
Gas payment, up to $3.00/MMBtu	up to $563k
Land lease at $1,000 to $2,000 per acre per year	$7k to $15k

The flared flow is falling: about 1,067 scfm in 2023 and 758 scfm in 2024, and the landfill had about 2 years of constructed capacity left at the end of 2024. The City's 2024 report gives 398,563,200 cubic feet to the enclosed flare over 8,328 hours, a full-year average of about 758 scfm; the engines took another 270,688,822 cubic feet (about 515 scfm). Only the flared gas is sized. Prices in US dollars.

WHAT IT WOULD LOOK LIKE

- Our own gas engines on a fenced parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- How fast the flared flow keeps falling (about 1,067 scfm in 2023, 758 in 2024) as the engines take more and waste placement ends
- 2025 flare and engine volumes (the 2025 NYSDEC report is not posted yet; EIA-923 shows the engine plant burned about a third less fuel in 2025)
- Gas outlook once waste placement ends: the 2024 report shows about 2 years of constructed capacity left at the permit rate
- Whether the gas lease with the engine plant covers all landfill gas or only what its engines take
- NYSDEC air permit under 6 NYCRR Part 201 and a tie-in under the landfill's Title V permit and its NSPS/NESHAP landfill gas rules
- Gas quality (siloxanes, H2S) for engines

Source: City of Albany, 2024 Solid Waste Management Facility annual report to NYSDEC (facility 01S02), Section 11: https://extapps.dec.ny.gov/fs/projects/SWMF/Annual%20Reports_Solid%20Waste%20Management%20Facility/Annual%20Reports_by%20Activity%20Type/Landfill/Landfill%20Annual%20Reports%20-%202024/R4/01S02_Albany_msw_R4_2024.2025-2-27.AR.pdf (seen 2026-11-13); 48.8% methane. The 2023 report gave about 1,067 scfm flared and 502 scfm to the engines. This is an estimate from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.